

$$V=\frac{1}{6}\parallel P_1-P_0,\;P_2-P_0,\;P_3-P_0\parallel$$

$$P_0,\ldots,P_3^{1/6}$$

$$V_{ivm}=\frac{1}{4}\left|\left(\begin{array}{ccccc}a_0&a_1&a_2&a_3&1\\b_0&b_1&b_2&b_3&1\\c_0&c_1&c_2&c_3&1\\d_0&d_1&d_2&d_3&1\\1&1&1&1&0\end{array}\right)\right|$$

$$M(q_0,q_1,q_2,q_3)=\left[\begin{array}{ccccc}q_{01}&q_{02}&q_{03}&q_{04}&1\\q_{11}&q_{12}&q_{13}&q_{14}&1\\q_{21}&q_{22}&q_{23}&q_{24}&1\\q_{31}&q_{32}&q_{33}&q_{34}&1\\1&1&1&1&0\end{array}\right],\qquad V_{ivm}=\frac{1}{4}\mid M(q_0,q_1,q_2,q_3)\mid$$

$$V_{xyz}=\frac{1}{6}\left|\begin{pmatrix}x_a&y_a&z_a&1\\x_b&y_b&z_b&1\\x_c&y_c&z_c&1\\x_d&y_d&z_d&1\end{pmatrix}\right|,\qquad V_{ivm}=S3\,V_{xyz},\quad S3=\sqrt{9\over 8}$$

$$S3=\sqrt{9/8}$$

$$_{288}V^2=\begin{pmatrix}0&1&1&1&1\\1&0&d_{01}^2&d_{02}^2&d_{03}^2\\1&d_{10}^2&0&d_{12}^2&d_{13}^2\\1&d_{20}^2&d_{21}^2&0&d_{23}^2\\1&d_{30}^2&d_{31}^2&d_{32}^2&0\end{pmatrix}$$

$$d_{ij}P_iP_j$$

$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$_{144}V_{xyz}^2=4a^2b^2c^2-a^2\left(b^2+c^2-f^2\right)^2-b^2\left(c^2+a^2-e^2\right)^2-c^2\left(a^2+b^2-d^2\right)^2+\left(b^2+c^2-f^2\right)\left(c^2+a^2-e^2\right)\left(a^2+b^2-d^2\right)$$

$$a,b,cd,e,f\\ V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$V_{ivm}=\frac{1}{4}\left|\begin{pmatrix}a_1-a_0&a_2-a_0&a_3-a_0\\b_1-b_0&b_2-b_0&b_3-b_0\\c_1-c_0&c_2-c_0&c_3-c_0\end{pmatrix}\right|$$

$$P_1-P_0P_2-P_0P_3-P_0$$

$$F_{i,j}=\mathbb{E}\left[\frac{\partial}{\partial\theta_i}\frac{p(x;\theta)}{p(x;\theta)}\frac{\partial}{\partial\theta_j}\frac{p(x;\theta)}{p(x;\theta)}\right]$$

$$F_{i,j} = \frac{1}{N} \sum_{n=1}^N \frac{\partial}{\partial \theta_i} p(x_n; \theta) \frac{\partial}{\partial \theta_j} p(x_n; \theta)$$

$$NN \rightarrow \infty$$

$$\theta \leftarrow \theta - \eta F(\theta)^{-1} \nabla_{\theta} L(\theta)$$

$$\mathcal{F} = -\, P(o \mid s) + [Q(s) \parallel P(s)]$$

$$q=(a,b,c,d)k=(a,b,c,d)q' = q-(k,k,k,k)$$

$$M\mathbb{R}^3\mathbb{R}^4d(q_1,q_2)=\|M(q_1)-M(q_2)\|_2$$

$$ds^2 = - c^2 \, dt^2 + dx^2 + dy^2 + dz^2$$

